

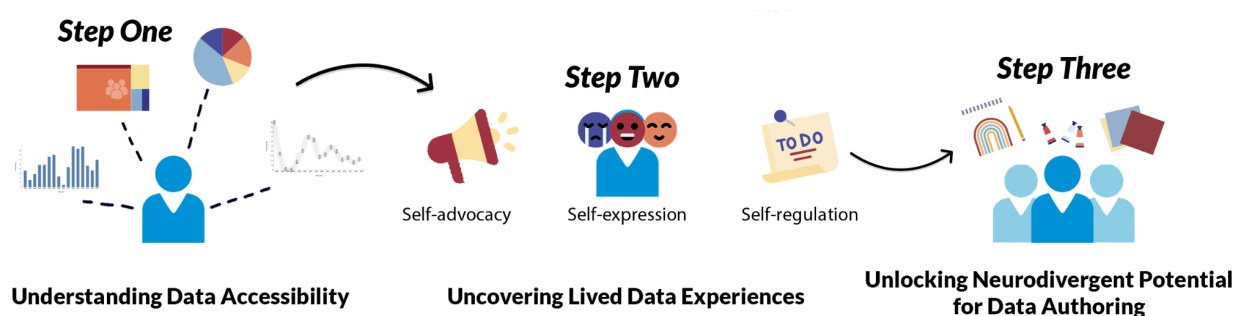


Figure 1. The path to cognitively accessible visualizations.

Visualization empowers people to access, understand, and use data for informed decision-making. Yet, conventional approaches often assume a one-size-fits-all user, neglecting individual differences and creating barriers to equitable access. My research seeks to dismantle these barriers by designing and developing visualizations that embrace individuality, integrate cultural relevance, and foster meaningful, engaging data experiences for diverse audiences.

To advance this vision, my past research focused on improving data accessibility for individuals with Intellectual and Developmental Disabilities (IDD), including conditions such as Down syndrome, Autism, and Cerebral Palsy. Through mixed-methods approaches, I developed accessible design guidelines, identified unique challenges and needs in data access for this population, and explored neurodivergent data expression and exploration practices through co-design. These efforts have established cognitive accessibility as a vital area in data visualization, driving inclusion and breaking down barriers for cognitively diverse populations.

During my PhD, I published in leading computer science venues such as IEEE VIS, ACM CHI, and ACM ASSETS, as well as disability-focused conferences like AUCD and AAIDD. My work on data accessibility [3] for individuals with IDD received a **Best Paper Award** at ACM CHI 2021, leading to an invited talk at the 2022 Grace Hopper Celebration. My co-design proposal with individuals with IDD won the Ray Hauser Award at the University of Colorado Boulder in 2021 and resulted in nominations for the Adobe Research Fellowship and the NCWIT Collegiate Award. In 2023, I was invited to the Schloss Dagstuhl seminar on Inclusive Visualization, and in 2024, I co-organized the inaugural Accessible Visualization workshop at IEEE VIS, contributing to a growing dialogue on inclusive design and data representation.

Building on this foundation, my postdoctoral research expands on these efforts to reach diverse audiences with varying abilities, data literacy levels, and unique characteristics. At the Cognition and Visualization Lab, I am engaged in various initiatives that deepen my expertise and broaden my research. Specifically, my work focuses on leveraging storytelling, aesthetics, and creative technologies to create engaging and impactful human-data interactions. By addressing real-world challenges, I aim to contribute to meaningful societal change.

Past Research

Initially inspired by a project aimed at supporting data-driven financial self-advocacy for individuals with IDD [1], my past research has focused on addressing data accessibility challenges related to diverse cognitive abilities, organized into three key thrusts detailed below.

(1) Understanding Data Accessibility

Neurodivergent individuals face significant challenges in intellectual functioning and often process abstract information differently from neurotypical populations. Conventional visualizations, often reliant on abstract encodings and minimalist design, can hinder their ability to interpret information. To assess the effectiveness of these visualizations and identify accessibility barriers, we conducted a graphical perception experiment [2, 3, 4] examining three design elements: chart type (e.g., bars, lines, pies), embellishments (e.g., chartjunk, icons), and continuity (e.g., continuous encodings like line charts versus discrete encodings like scatterplots).

Our findings revealed that individuals with IDD employ distinct strategies to interpret data, yet conventional designs often fail to support these approaches, resulting in exclusion. For instance, pie charts posed significant challenges for interpreting proportions, highlighting the limitations of conventional visualizations. In contrast, visual metaphors improved comprehension, while discrete encodings improved working memory, challenging minimalist design principles. Additionally, the prevalence of cognitive and sensory challenges in this population underscores the need for sensory-sensitive approaches to data interaction, extending design considerations beyond the visual.

From these insights, we developed four design guidelines (Figure 2.), emphasizing the need for innovative approaches to create cognitively accessible visualizations.

(2) Uncovering Everyday Data Experiences

A lack of understanding of how individuals engage with data in daily life hinders the design of effective and relatable visualizations. To address this, we conducted semi-structured interviews with 15 individuals in the IDD community, focusing on three key areas: their definitions and perceptions of data, scenarios of data use, and the barriers they face in accessing it.

We found that individuals with IDD often use personal data to navigate their daily lives, yet most of these interactions were unconscious and unsupported by conventional visualizations. We identified three primary data scenarios: (1) using data for self-advocacy and storytelling (e.g., demonstrating competence or advocating for independence), (2) for self-expression and compassion (e.g., in psychiatric treatments or fostering social understanding), and (3) for self-regulation and autonomy (e.g., managing personal health and daily functioning).

These findings [5, 6] reimaged the role of visualization in data accessibility, underscoring the importance of approachable designs that make data more relevant and engaging, while also showcasing visualization's potential in therapeutic analytics and inclusive data education.

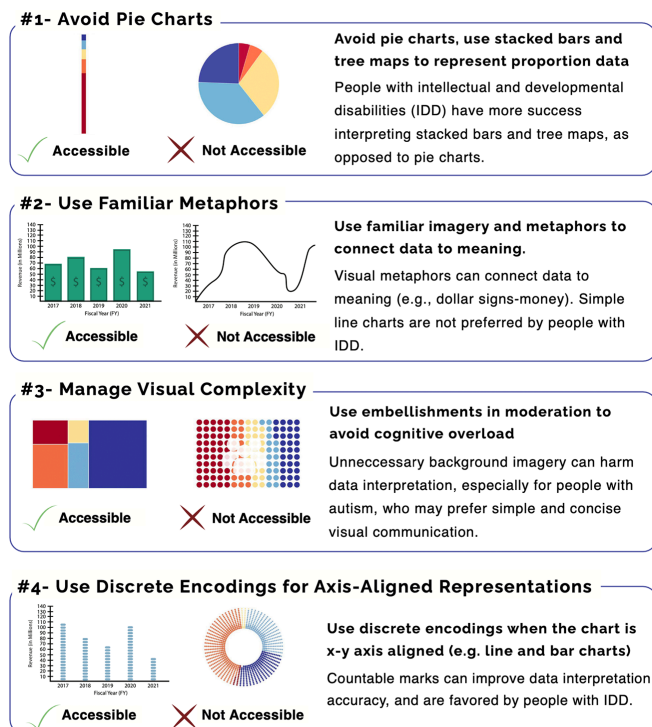


Figure 2. Design guidelines derived from the experiment.

(3) Unlocking Neurodiversity for Data Authoring

Conventional visualizations are often designed without input from neurodivergent individuals, overlooking their unique needs in data expression and authoring. To bridge this gap and develop culturally relevant solutions, we partnered with 20 individuals with IDD as co-designers, collaboratively creating visualizations that reflect their lived experiences and generating cognitively inspired strategies for more inclusive visualizations.

Through hybrid participatory design workshops, we identified three distinct strategies individuals with IDD use to engage with data and create visualizations: hands-on exploration, personal narratives, and visual metaphors and analogies. These approaches reflect their unique ways of understanding data and their inclination to translate it into concrete, meaningful representations.

This process [8] offered valuable insights into their mental models, deepening our understanding of cognition and its connection to visualization. It led us to propose two alternative guidelines for designing cognitively accessible visualizations: (1) translating numbers into narratives and (2) blending data design with everyday aesthetics. These findings have important implications for accessible data representation, interaction, education, and communication.

Future Directions

Overall, my past research efforts have established a foundation for rethinking conventional visualization practices and highlighted the importance of inclusive design in fostering equitable human-data interaction. Building on these insights, my future work aims to engage broader audiences, explore diverse contexts, and integrate innovative technologies. I am excited to collaborate with faculty across disciplines, leveraging our shared passion to advance impactful and inclusive research in **equitable data interaction**.

(1) Amplifying Voices Through Data-Driven Storytelling

Data has the power to tell stories that inspire empathy, advocacy, and action. However, many individuals—especially those from underrepresented or underserved communities—lack access to the tools or resources needed to craft their own narratives. Building on my ongoing work with the *Data Poetics* event for the Atlanta Science Festival and *Accessible Data Storytelling* for people with IDD, I aim to explore how data storytelling can amplify diverse voices and foster social change. This agenda includes designing innovative visualization templates, prototyping intelligent and accessible authoring tools, and developing inclusive research methods that empower individuals to actively shape and share their stories. By democratizing data storytelling, my work seeks to make data a more inclusive and transformative medium for advocacy and representation.

(2) Harnessing Visualization for Therapeutic Expression

Visualization has immense potential as a therapeutic tool for managing emotions, fostering self-awareness, and building resilience. By developing novel visual analytics systems for personal data management, I aim to explore how visualization can support individuals in understanding and processing their emotions. This work will examine the integration of visualization into therapeutic practices such as emotion regulation and cognitive-behavioral therapy. Future directions include conducting empirical studies to assess the impact of design elements on emotional responses, creating tools that enable self-expression through personal data, and developing interaction techniques to support collaboration between individuals and mental health professionals. By broadening the scope of therapeutic applications, this research seeks to make visualization a meaningful tool for emotional well-being and mental health care.

(3) Advancing Embodied Learning for Data Reasoning

Cognitive and sensory differences significantly influence how individuals understand and reason with data. Embodied cognition provides a powerful framework for designing accessible and engaging data interactions. In my current research, I am exploring how technologies like augmented and virtual reality, combined with tactile-kinesthetic and multisensory approaches, can enhance data literacy, foster autonomy, and support everyday decision-making. Through this work, I am developing immersive environments that deepen engagement with data and make reasoning processes more intuitive for diverse users. These efforts demonstrate my commitment to creating adaptable systems that address a broad range of cognitive and sensory needs, advancing the inclusivity and impact of data reasoning.

Conclusion

In summary, my research is driven by a vision to **make data more inclusive, accessible, and meaningful** for diverse populations, particularly those underserved by conventional visualization designs [7]. By exploring the intersections of storytelling, aesthetics, and emerging technologies, I aim to address real-world challenges and contribute to broader social good. I am excited to continue building on this foundation, expanding interdisciplinary collaborations, and exploring new frontiers in data accessibility and human-data interaction. Together, I believe we can redefine how data is understood, used, and shared to create a more just and equitable society.

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